

SANTA MARIA, Brazil

WMO: 839370

Lat: 29.711S

Lon: 53.688W

Elev: 287

StdP: 14.54

Time Zone: -3.00 (SAB)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.0	39.2	31.8	26.5	43.2	34.7	29.7	43.8	23.2	75.5	19.9	74.7	1.6	120	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.7	96.8	75.8	93.6	75.2	91.3	75.0	80.3	89.3	79.0	88.0	77.8	86.4	8.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.2	147.9	85.4	76.8	140.9	84.0	75.3	134.0	82.7	44.2	90.1	42.7	88.0	41.4	86.4	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	16.2	14.0	30.5	101.9	2.1	1.5	29.0	103.0	27.7	103.9	26.5	104.7	25.0	105.8
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.2	78.1	77.9	73.9	68.9	62.9	58.4	57.1	61.3	63.6	68.0	72.4	76.4
	DBStd	9.94	4.91	5.19	5.61	6.51	6.93	8.49	8.97	9.16	7.43	5.97	5.52	5.16
	HDD50	56	0	0	0	0	3	16	27	9	1	0	0	0
	HDD65	1023	0	0	5	35	123	238	281	186	112	34	8	1
	CDD50	6695	872	781	741	567	401	268	247	360	409	559	672	818
	CDD65	2188	408	361	281	153	56	39	36	72	70	128	230	354
	CDH74	18722	3744	3190	2150	1070	349	259	273	713	603	946	2058	3367
	CDH80	7381	1627	1315	806	365	87	49	43	228	236	329	790	1506
(14) Wind	WSAvg	5.9	6.2	5.7	5.3	4.6	4.8	5.1	5.6	6.5	6.8	6.9	6.8	6.3
(15) Precipitation	PrecAvg	70.5	6.4	5.9	5.0	5.9	5.0	5.2	6.0	4.3	6.7	8.0	5.6	6.7
	PrecMax	107.5	17.4	11.6	11.8	13.5	11.3	10.4	10.4	10.4	13.7	17.0	17.4	19.3
	PrecMin	43.8	2.5	1.8	0.6	1.3	0.3	0.2	2.0	0.8	2.2	2.2	1.0	0.0
	PrecStd	14.1	4.0	2.7	2.9	2.8	2.6	2.3	2.8	2.1	3.1	3.7	3.8	4.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.8	99.0	98.4	93.1	87.6	84.5	84.2	89.7	93.5	95.4	97.2	99.2
		MCWB	76.4	77.0	76.0	71.7	71.4	69.6	66.8	69.8	70.2	71.9	71.4	75.0
	2%	DB	95.3	94.7	92.3	88.1	82.4	80.8	80.8	86.0	86.4	89.2	91.8	94.9
		MCWB	76.9	78.8	76.5	72.2	68.7	67.8	65.8	68.6	68.6	72.2	72.1	74.0
	5%	DB	92.6	91.0	88.1	84.3	77.8	76.9	77.1	82.1	80.6	82.5	88.0	91.4
		MCWB	77.1	77.7	75.5	71.3	67.9	66.3	64.7	66.9	67.9	69.5	70.3	73.8
	10%	DB	89.2	87.7	84.3	80.5	73.4	71.7	71.6	77.3	75.4	78.9	84.4	88.1
		MCWB	75.9	76.7	73.8	69.9	66.3	65.0	63.5	65.6	65.9	68.1	69.5	73.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.4	82.3	82.0	76.8	74.3	71.4	69.6	72.5	74.0	76.2	77.2	80.0
		MCDB	93.2	89.9	89.3	86.3	81.2	80.5	78.3	84.5	85.0	89.1	87.6	89.7
	2%	WB	80.0	80.3	79.0	74.6	71.2	69.3	67.4	69.8	71.1	73.8	74.7	78.5
		MCDB	89.7	88.8	87.2	83.4	77.7	77.6	77.1	81.9	81.3	84.8	84.5	88.6
	5%	WB	78.6	79.0	77.1	72.9	69.2	67.3	65.6	68.0	69.2	71.6	73.3	76.9
		MCDB	88.0	87.1	84.7	80.3	75.2	74.8	74.7	78.9	78.0	79.6	82.8	86.6
	10%	WB	77.2	77.6	75.4	71.3	67.4	65.3	63.8	66.2	67.3	69.7	71.8	75.2
		MCDB	85.5	85.3	82.3	77.8	72.4	71.8	70.8	75.9	74.5	76.6	81.0	84.7
(35) Mean Daily Temperature Range	5% DB	MDBR	18.7	18.0	19.0	19.8	17.1	16.8	18.1	19.8	18.2	19.0	21.4	20.8
		MCDBR	24.0	21.4	23.6	24.7	22.7	20.7	23.4	25.0	26.5	25.2	27.3	26.2
		MCWBR	9.2	9.4	10.4	10.9	11.8	11.0	12.3	11.6	12.8	11.8	11.3	10.4
	5% WB	MCDBR	19.8	18.8	20.4	19.9	18.3	17.8	20.6	23.1	21.8	21.8	21.7	21.9
		MCWBR	9.1	9.4	10.7	9.7	10.6	10.0	11.9	12.3	11.7	11.3	10.7	10.2
(40) Clear-Sky Solar Irradiance	taub		0.422	0.413	0.399	0.393	0.365	0.388	0.390	0.490	0.519	0.460	0.404	0.424
	taud		2.342	2.372	2.400	2.382	2.433	2.360	2.326	1.992	1.937	2.158	2.338	2.309
	Ebn at Noon		292	289	282	267	259	242	247	231	242	272	295	292
	Edh at Noon		43	40	37	34	29	30	32	49	58	49	43	44
(44) All-Sky Solar Radiation	RadAvg		2084	1918	1630	1267	930	775	861	1086	1314	1621	2033	2143
	RadStd		174	119	126	123	99	67	98	95	133	205	242	195

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page